

FAHAD RAHMAN CH

BCE210

1) string= "Hello I am niveditha rejin from tsr" print the largest length word ?

Ans :

```
const word = "Hello I am niveditha rejin from tsr";
const string = word.split(" ");
const longest = string.reduce((long,current) => {
    return long.length>current.length ? long : current;
},0)
console.log(longest);
```

2) string=" welcome to java script" to print vowels?

Ans)

```
const ar=['a','e','i','o','u'];
const string=" welcome to java script";
const split=string.split("");
for(let i=0;i<split.length;i++){
    for(let j=0;j<ar.length;j++){
        if(split[i]==ar[j]){
            console.log(split[i])
        }
    }
}
```

3)reduce odd number sum of from array?

Ans:

```
let arr=[1,2,3,4,5];
let arrn=arr.reduce((acc,curr)=>{
```

```
    if(curr%2==1){  
        return acc+curr  
    }  
    return acc  
},0)  
console.log(arrn)
```

4) object and array destructuring ?

Array destructuring

Ans:

```
let array=[1,2,3,4,];  
const[first,second,third,fourth]=array;  
console.log(first)  
console.log(second)  
console.log(third)  
console.log(fourth)
```

object destructuring

```
let arr={  
    name:'fahad',  
    age:18,  
}  
const{name,age}=arr  
console.log(name)  
console.log(age)
```

5) Take element from the array which is divide by 5 and takes its sum ?

Ans:

```
let arr=[1,2,3,4,5,6,7,10];  
let fil=arr.filter(item=>item%5==0);  
let reduce=fil.reduce((acc,curr)=>acc+curr);  
console.log("sum : "+reduce)
```

6)ooobject method seal,freeze ?

Ans:

freeze

```
let user={  
  name:'fahad',  
  age:18,  
  place:'wandoor'  
}  
Object.freeze(user);  
user.name='rahman'  
console.log(user)
```

seal

Ans:

```
let user={  
  name:'fahad',  
  age:18,  
  place:'wandoor'  
}  
Object.seal(user);
```

```
user.name='rahman'
```

```
console.log(user)
```

6) find out second largest

Ans:

```
let array=[1,2,3,4];
```

```
let double=array.concat(array);
```

```
console.log(double);
```

```
let max=Math.max(...double);
```

```
console.log("largest number in array : "+max)
```

```
let secmax=Math.max(...double.filter(num=>num!==max))
```

```
console.log("second largest num in an array : "+secmax)
```

7)optional chaining ?

```
let user={
```

```
  name:'fahad',
```

```
  age:18,
```

```
  address:{
```

```
    place:'wandoor'
```

```
  }
```

```
}
```

```
console.log(user?.address?.place)
```

8) Genarator function return an id infinityly its called ?

Ans

```
function*fun(n){  
  for(i=1;i<=n;i++){  
    yield 5*i  
  }  
}  
  
let hi=fun(10)  
let id=setInterval(()=>{  
  res=hi.next()  
  
  if(res.done===true){  
    clearInterval(id)  
  }else{  
    console.log(res.value)  
  }  
},1000)
```

9)how to check the object is empty or not

Ans

```
function hi(obj){  
  return Object.keys(obj).length==0;  
}  
  
let emptyobject={};  
let nonobject={key:'hi'}  
console.log(hi(emptyobject));  
console.log(hi(nonobject));
```

10) (a,e,i,o,u) remove in an string;

Another method

```
let a=['a','e','o','i','u'];
let an='hlo my name is fahad'
let arrn=an.split("");
for(i=0;i<arrn.length;i++){
    if(a.includes(arrn[i])){
        console.log(arrn[i])
    }
}
```

Another method

```
let a=['a','e','i','o','u'];
let b='my name is fahad';
let c=b.split("");
for(let key of c){
    if(a.includes(key)){
        console.log(key)
    }
}
```

11) Remove the duplicate element in the array?

```
let a=[1,1,2,2,3];
n=a.length
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
```

```

    if(a[i]==a[j]&&i!=j){
        for(k=i;k<n;k++){
            a[k]=a[k+1];
        }
        i--;
        n--;
    }
}

```

a.length=n

console.log(a)

Another method

```
let arr=[1,2,2,3,3,4];
```

```
let uniq = [ ...new Set(arr)];
```

```
console.log(uniq)
```

another method

```
let user=[1,2,3,3,3];
```

```
let uni=user.filter((item,index)=>user.indexOf(item)===index);
```

```
console.log(uni)
```

12) Remove the duplicates elements in string ?

```
let s='fahad';
```

```
let a=s.split("");
```

```
let c=[...new Set(a)]
console.log(c)
13 )trim();
let a=' fahad ';
console.log(a.trim())
```

13)average of the array ?

```
let arr=[1,2,3,4,5,6];
let sum=arr.reduce((acc,curr)=>(acc+curr),0);
let avg=sum/arr.length
console.log("avarage of the array : "+avg)
```

14) find first large second large element in an array ?

```
let ar=[1,23,4,5];
let max=Math.max(...ar);
let secmax=Math.max(...ar.filter(num=>num!=max));
console.log(secmax)
console.log(max);
```

15)promise all

```
let promise1=new Promise((resolve,reject)=>{
    resolve([1,2]);
})
let promise2=new Promise((resolve,reject)=>{
    resolve([3,4]);
});
```



```
const allpromise =Promise.all((promise1,promise2))
```

```
allpromise
```

```
.then((res)=>{  
    console.log(res)  
})  
.catch((err)=>{  
    console.log(err)  
});
```

```
16)promise.allsettled
```

```
let promise1=new Promise((resolve,reject)=>{  
    resolve("success");  
})  
let promise2=new Promise((resolve,reject)=>{  
    resolve("fail");  
});  
const allpromise =Promise.allSettled([promise1,promise2])
```

```
allpromise
```

```
.then((res)=>{  
    console.log(res)  
})  
.catch((err)=>{  
    console.log(err)  
});
```

17) promise.race

```
let promise1=new Promise((resolve,reject)=>{
    resolve("fail");
})
let promise2=new Promise((resolve,reject)=>{

    resolve("success");
});
const allpromise =Promise.race([promise1,promise2])
```

allpromise

```
.then((res)=>{
    console.log(res)
})
.catch((err)=>{
    console.log(err)
});
```

18)promise.any()

```
let promise1=new Promise((resolve,reject)=>{
    reject("succes");
})
let promise2=new Promise((resolve,reject)=>{
    resolve("fail");
});
const allpromise =Promise.any([promise1,promise2])
```

```
allpromise
.then((res)=>{
  console.log(res)
})
.catch((err)=>{
  console.log(err)
});
```

19) Find highest odd number in the array ?

```
let arr=[1,2,3,7,4,5];
let arrn=arr.reduce((acc,curr)=>{
  if(curr%2==1&&curr>acc){
    return curr
  }
  return acc;
},0)
console.log(arrn)
```

19) call ?

```
let obj={
  name:'sanitta',
  data:function(){
```

```
        console.log(this.name)
    }
};
let obj2={name:'sanu'}
obj.data.call(obj2)
```

another example:

```
let obj={
    name:'fahad',
    age:18,
    hi:function(){
        console.log(this.name)
    }
}
let obj1={
    name:'milu',
    age:19,
}
obj.hi.call(obj1)
```

20) apply()

```
let arr={
    name:'fahad',
    age:18,
```

```
}  
function hi(n,n1){  
    console.log(n,n1+this.name)  
}  
hi.apply(arr,['hi','hlo'])
```

21) Even numbers to find square and odd numbers to find cube ?

```
let arr=[1,2,3,4,5];  
let arrn=arr.map(item=>{  
    if(item%2==0){  
        return Math.pow(item,2)  
    }else{  
        return Math.pow(item,3)  
    }  
})  
console.log(arrn)
```

22) Remove even numbers from the array ?

```
let arr=[1,2,3,4,5];  
let arrn=arr.map(item=>{  
    if(item%2==0){  
        return Math.pow(item,2)
```

```
    }else{  
        return Math.pow(item,3)  
    }  
})  
console.log(arrn);
```

23) [{a:13},{b:14},{c:43}]; splice(0,1)

```
let arr=[{a:13},{b:45},{c:65}];  
let arrn=arr.splice(0,1)  
console.log(arr)
```

24) reduce smallest odd number element in an array ?

```
let arr=[1,2,9,4,5,6,7,8];  
let small=arr.reduce((acc,curr)=>{  
    if(curr%2==1&&curr<acc){  
        return curr  
    }  
    return acc  
},Infinity);  
console.log(small)
```

25) currying ?

```
function sum(a){  
  return function(b){  
    return function(c){  
      return a+b+c;  
    }  
  }  
}  
  
console.log(sum(10)(40)(80))
```

26) class and 2 instance ?

```
class A{  
  constructor(a){  
    console.log(a)  
  }  
  hi(b){  
    console.log(b)  
  }  
}  
  
let car=new A(1);  
let cars=new A(9);  
car.hi(90);  
cars.hi(60);
```

27) multiple of 2

```
let i=0;
function hi(a){
  while(i<=10){
    console.log(i*a)
    i++;
  }
}

hi(2);
```

```
28) flat();
let a=[1,2,[3,4],[6,7],7,8];
let an=a.flat();
console.log(an);
```

29) call back function ?

```
function hi(h,callback){
  console.log(h)
  callback();
}

function a(){
  console.log('hello');
}

hi(10,a);
```


30) A array of numbers which contain positive and negative integers and find the positive sum of even negeative integers.

```
let arr=[10,-13,-4,5,7,-8,-6,19];
```

```
let an=arr.filter((item)=>{
```

```
    if(item<0){
```

```
        return item
```

```
    }
```

```
})
```

```
let even=an.map((item)=>{
```

```
    if(item<0){
```

```
        return item*-1
```

```
    }
```

```
})
```

```
console.log(even);
```

```
let sum=even.reduce((acc,curr)=>{
```

```
    if(curr%2==0){
```

```
        return curr+acc
```

```
    }
```

```
    return acc
```

```
},0)
```

```
console.log(sum)
```

another method

```
let arr=[10,-13,-4,5,7,-8,-6,19];
```

```
let sum=arr.filter((item)=>{
```

```
    if(item<0){
```

```
        return item
```

```

    }
  }).map((item)=>{
    if(item<0){
      return item*-1
    }
  }).reduce((acc,curr)=>{
    if(curr%2==0){
      return curr+acc
    }
    return acc
  },0)

```

console.log(sum)

31) To print the duplicate odd number in an array

```
let a=[1,1,1,2,2,3,3,9,9];
```

```
let arr=[];
```

```
for (let i = 0; i < a.length; i++) {
```

```
  for (let j = 0; j < a.length; j++){
```

```
    if(i!=j&&a[i]%2==1){
```

```
      if(a[i]==a[j]){
```

```
        arr.push(a[i])
```

```
      }
```

```
    }
```

```
  }
```

```
}
```

```
console.log(arr)
```

```
let c=[...new Set(arr)]  
console.log(c)
```

32) Promise , async , await ?

```
let promise= new Promise((resolve,reject)=>{  
  let task = true;  
  if (task){  
    resolve("success");  
  }else{  
    reject("failed")  
  }  
})
```

```
async function call(){  
  try{  
    let data= await promise
```

```
        console.log(data)
    }
    catch(err){
        console.log(err)
    }
}
```

```
call();
```

33) object highest value keys

```
let user={
    a:18,
    b:89,
    c:90,
    d:88,
    e:90

}

let max=Math.max(...Object.values(user))
let keymax=Object.keys(user).filter(key=>user[key]==max);
console.log(keymax)
```

34)sum odd numbers in object in an array ?

```
let arr = [{a:10,b:1},{a:11,b:1},{a:9,b:10}];
```

```
let rd=arr.reduce((acc,curr)=>{
```

```
    let sum=0;
```

```
    for(key in curr){
```

```
        if(curr[key]%2==1){
```

```
            sum=sum+curr[key];
```

```
        }
```

```
    }
```

```
    return acc+sum
```

```
},0)
```

```
console.log(rd)
```

```
35) bind();
```

```
let user={
```

```
    name:'fahad',
```

```
    age:19,
```

```
}
```

```
function hi(){
```

```
    console.log("hello"+this.name)
```

```
}
```

```
let gy=hi.bind(user)
```

```
gy()
```

```
gy()
```

```
gy()
```

35) timing features ?

```
function hi(){  
  for(i=0;i<n;i++){  
    console.log("hello")  
  }  
}
```

```
let id =setTimeout(()=>{  
  console.log("hello")  
},1000)
```

```
setTimeout(()=>{  
  clearTimeout(id)  
  console.log("stop")  
},3000)
```

36) string ="hello my name is fahadch" print the largest length word ?

```
let hi="hello my name is dhyaaan ";
```

```
let a=hi.split(' ');
```

```
let hii=a.reduce((acc,curr)=>{
```

```
  if(curr.length>acc.length){
```

```
    return curr
```

```
  }
```

```
  return acc
```

```
}, "")
```

```
console.log(hii)
```

37) 'a','e','i','o','u', remove the vowels in string?

```
let a=['a','e','i','o','u'];
```

```
let ar='fahadrahman'
```

```

let an=ar.split("");
for(i=0;i<an.length;i++){
  for(j=0;j<a.length;j++){
    if(an[i]==a[j]){
      for(k=i;k<an.length;k++){
        an[k]=an[k+1];

      }
      i--;
      an.length--
    }
  }
}
console.log(an.join(""))

```

38)find freq in an string ?

```

let arr = 'strinnng';
let arrn = arr.split("");
let freq = {};
for(let i = 0; i < arr.length; i++){
  if(arr[i] in freq){
    freq[arr[i]]++;
  }
  else{
    freq[arr[i]] = 1;
  }
}

```

```
}
```

```
console.log(freq);
```